

Akash Kothari

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering (Co-op)

Waterloo, ON

GPA: 3.9/4.0

EXPERIENCE

AI/ML Powered Software Engineering Intern

Jan. 2026 - Present

RamSoft

Toronto, ON

- Reduced transcription **word error rate by 80%** across 200 clinical audio samples by building a **MedASR** pipeline with chunked streaming, PyTorch word-confidence scoring, and a chunk-replay simulator for iterative evaluation.
- Designed a **Cosmos DB** → **SQL Server** schema migration for a scheduling epic, producing normalized **AppointmentV2** and **Reservation** table definitions with wall-clock time storage and full recurrence support.
- Built an internal **MCP server** for **Azure SQL** with MFA token auth, enabling natural-language database queries

Cloud Software Engineering Intern

Jun. 2025 - Sep. 2025

Tripleview Technologies

Mississauga, ON

- Automated Teams call recording retrieval using **Azure**, **Microsoft Graph**, and **C#**, eliminating **10+ hours/week** of manual work for customer support teams.
- Built an **AI summarization pipeline** with webhook triggers and polling logic, processing **8+ transcripts/hour** into structured **Zendesk** tickets and reducing team miscommunication.
- Owned the full dev lifecycle delivering **production Azure tooling** now expanded across internal teams.

PROJECTS

FocusOS - Personal Productivity OS | *FastAPI, PostgreSQL, React, Azure OpenAI, Docker*

- Built a full-stack productivity OS with an **AI job analysis pipeline** using **Azure OpenAI** — scores resume-job match, extracts ATS keywords, and generates tailored LaTeX resumes.
- Designed a **multi-step background task pipeline** with non-blocking AI calls, retry logic, and real-time status polling, deployed with **Docker** and a **PostgreSQL** backend.
- Surfaced real-time productivity insights via a normalized **multi-table PostgreSQL** schema and **React** site

Mercury Mortgages — AI Expense Management System | *FastAPI, Microsoft Graph, PostgreSQL, React*

- Built a full-stack expense management system for a local business processing **100+ receipts/week** across 5 corporate cards, automating receipt matching and exportable expense reports, **eliminating a manual role**.
- Engineered a **dual-ingestion pipeline**, manual upload and email parsing via **Microsoft Graph** webhooks, with async background processing using threaded PDF conversion and **Azure OpenAI** vision for field extraction.
- Implemented a **rules engine** mapping extracted vendor and location data to GL codes and sub-companies, with **role-based access control** and admin-gated expense report approvals and snapshots.

Mathora - Learn Math Visually | *React Three Fiber, Three.js, LLMs, Express*

- Built an LLM-controlled math engine driving **15+ animation types** across 3D graphing, whiteboard, camera, and voiceover via structured prompt schemas, with **10 precomputed templates** for instant rendering.
- Designed a **custom animation library** with a parameter schema exposed to the LLM, enabling deterministic visual execution with safe fallbacks.
- Reduced perceived latency from **30s to under 2s** by streaming animation steps via **SSE** as they were generated

Yuno Ball - NBA Odds and Prediction Engine | *Python, Flask, React, XGBoost, Data Engineering*

- Scraped + processed **9,000+ NBA games**, engineered **27** statistical features, and cut noise via feature ablation.
- Trained an **XGBoost model** achieving **64.8% accuracy**, benchmarked against **Polymarket odds**
- Built a full-stack system with **Flask API** and **React** frontend featuring graded bets, parlay builder, and a Gemini chatbot explaining decisions via feature importance.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Java, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Next.js, FastAPI, Flask, Node.js/Express, Three.js, React Three Fiber, GraphQL

Cloud & DevOps: Azure, Kubernetes, Docker, PostgreSQL, CI/CD, Terraform, REST APIs, Git, Jira, Confluence

ML/Data: Scikit-learn, PyTorch, XGBoost, Pandas, Feature Engineering, Computer Vision, NumPy, SciPy

Game/3D: Unity, WebGL